

Exercises Architecture

1. Name the main components of the M0 CPU and explain their function.

2. A processor executes a list of instructions in a predetermined order. Find analogies from everyday life.

3. How many memory bytes can be addressed with an 8-bit memory address? How many with a 16-bit address? How many with a 32-bit address?

4. Name 3 instruction types of the M0 CPU.

5.

6. What is the function of the following registers?
 - a. pc

b. sp

c. lr

7. Why is the PC initialized to a defined value at reset? (Although other CPU registers may have an undefined content.)

8. Name the different parts of a line in assembly code.

9. What is a memory map? What is it used for?

10. How many byte positions can be addressed by the M0? Which positions in the memory map need to be occupied? Explain your answer.

11. Explain the following terms

a. Fetch

b. Execute

c. Word

d. Half-word

e. Little endian

f. Big endian

g. Word Alignment

12. A program (code in C) has variables represented as below in the memory map. Determine the decimal values of the variables Var1 Var5. Assume Little Endian representation.

Address	Byte content (decimal, hex, binary)	Variable
0x2FFF'FFF7	0x45	
0x2FFF'FFF8	0xE2	Var4 (16-bit short)
0x2FFF'FFF9	01100010 (binary)	
0x2FFF'FFFA	213	
0x2FFF'FFFB	25	
0x2FFF'FFFC	0x65	Var3 (32-bit unsigned integer)
0x2FFF'FFFD	10101101(binary)	
0x2FFF'FFFE	0xA3	
0x2FFF'FFFF	0x82	
0x3000'0000	0xA2	Var5 (32-bit integer)
0x3000'0001	34	
0x3000'0002	0x54	
0x3000'0003	0xFF	
0x3000'0004	0x92	Var2 (unsigned char)
0x3000'0005	0x03	Var1 (char)

Var1 =

Var2 =

Var3 =

Var4 =

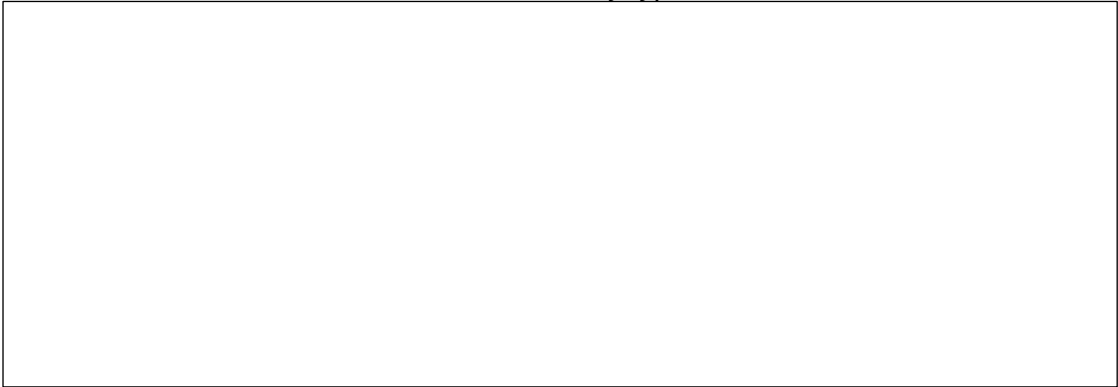
Var5 =

How would the same variable values be stored on a Big Endian platform? Fill in the table.

Address	Byte content (decimal, hex, binary)	Variable
0x2FFF'FFF7		
0x2FFF'FFF8		

0x2FFF'FFF9		
0x2FFF'FFFA		
0x2FFF'FFFB		
0x2FFF'FFFC		
0x2FFF'FFFD		
0x2FFF'FFFE		
0x2FFF'FFFF		
0x3000'0000		
0x3000'0001		
0x3000'0002		
0x3000'0003		
0x3000'0004		
0x3000'0005		

13. Which 3 memory areas (sections) can be differentiated for a program? What will be stored in the individual areas? In which memory type can the areas be located?



14. Assume that the following memory areas are used when executing a program:

- Code: 0x20000000 to 0x200001FF
- Data: 0x20000200 to 0x200002FF
- Stack: 0x20000300 to 0x200003FF

Draw an appropriate memory map and draw the three sections. Label the first and last addresses for each area. How many storage locations does each of the areas contain?

